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Implementation and performance of post-quantum cryptography for resource constrained consumer electronics

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Abstract
The proliferation of consumer electronics (CE), from smartphones to IoT-enabled smart devices, has significantly transformed our everyday lives, enhancing convenience and connectivity. However, these transformations to modern life have introduced security vulnerabilities, particularly as traditional cryptographic algorithms face mounting challenges from quantum computing technology. Post-quantum cryptography (PQC) has emerged as a promising solution for providing future-proof security for next-generation CE against the quantum computing threat. PQC algorithms are developed to resist cyberattacks from both classical and quantum computers, ensuring long-term security solutions. This article explores the integration of PQC into next-generation CE devices to ensure confidentiality, integrity, and user privacy. It highlights the vulnerabilities of classical cryptographic methods and underscores the need for transitioning to quantum-resistant solutions. Moreover, the practical implementation of PQC in resource-constrained CE devices is analyzed. Our evaluation shows that optimized PQC incurs negligible overhead on servers (5%), can be up to 12× slower on CE devices, and reduces communication overhead below conventional ECC (1 KB vs. 2.1 KB), while achieving hardware throughput thousands of times higher than RSA. By proactively exploring the opportunities and challenges associated with PQC, this research outlines future directions for implementing robust, scalable, and secure systems, thereby ensuring consumer trust in the evolving technological landscape.

Keywords Consumer electronics, Edge computing, Security, Post-quantum cryptography, Implementation, Standardization

1 Introduction

The integration of the Internet of Things (IoT) and fifth-generation (5G) wireless networks into traditional CE devices has transformed them into next-generation CE devices with ubiquitous connectivity, high intelligence, and control automation [1]. This shift enables users to experience a more interconnected, efficient and convenient lifestyle [2]. Consumers can now enjoy uninterrupted streaming, remote access to smart home appliances, and real-time updates on health and fitness metrics, no matter their location.

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Fig. 1 illustrates different categories of next-generation CE across different applications. With the ubiquitous connectivity of these devices and users' demands for more features and emerging technologies in CE devices, maintaining a balance between cost, features, and security is challenging for manufacturers [3]. These devices are being shaped by a diverse array of emerging technologies that cater to a wide range of applications [4]. These include advancements in 6G and 5G connectivity, IoT, autonomous vehicles, drones, robotics, virtual and augmented reality (VR/AR),intelligent data classification [5, 6], cybersecurity solutions, and artificial intelligence (AI), among others. While some of these technologies overlap with innovations developed for other industries, they transition into the consumer space when specifically adapted for user-friendly, practical applications. For instance, features initially designed for industrial or enterprise use may be tailored to meet the needs of everyday consumers, making them integral to the evolving landscape of CE. This cross-pollination of technology ensures that consumer devices remain at the forefront of functionality, convenience, and innovation. Additionally, with the rapid development of quantum computing technology, much like the advancements in next-generation CE, quantum computers are expected to become available soon, which will pose a significant threat to the security and privacy of these devices, potentially compromising traditional cryptographic methods.

Quantum computers can perform complex mathematical and intensive computational tasks exponentially faster than classical computers [7, 8]. It can solve complex science problems, which are even difficult to be solved by present-day supercomputers. On the one hand, these machines will further revolutionize AI technology with a massive boost in computational power. On the other side, it challenges the security assumptions underlying traditional cryptographic algorithms, sparking concerns about the susceptibility

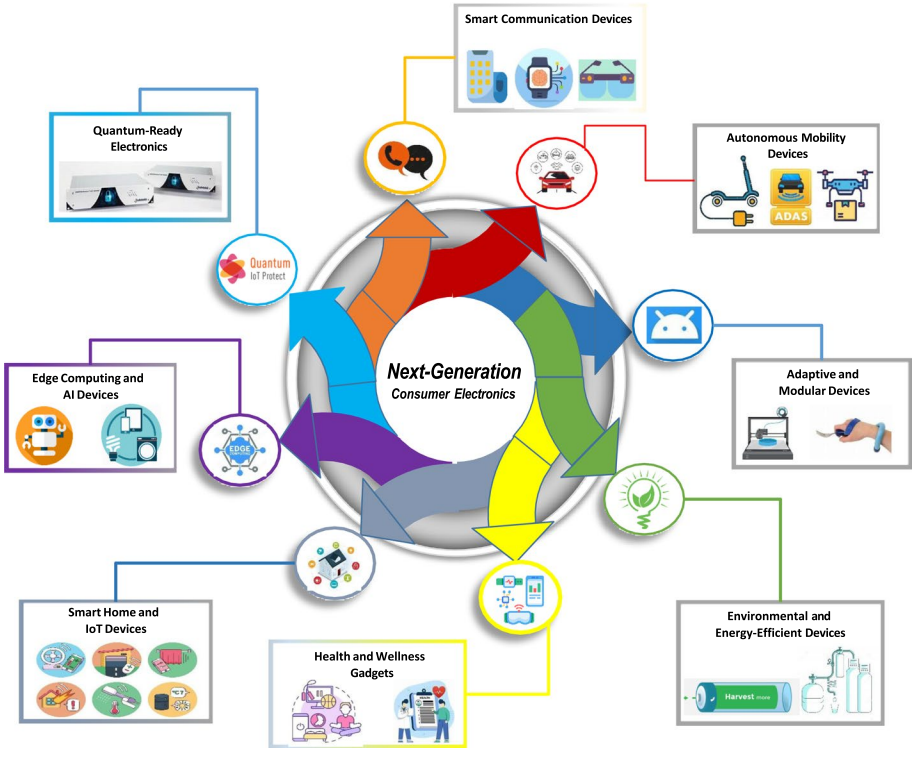

Fig. 1 Illustration of the different categories of next-generation consumer electronics devices across various applications

of sensitive data to quantum-powered attacks [9, 10]. The most popular asymmetric schemes, including Rivest, Shamir, Adleman (RSA) [11, 12], elliptic curve digital signature algorithm (ECDSA) [13–15], elliptic curve Diffie–Hellman (ECDH) [16] or digital signature algorithm (DSA) [17] has already broken in polynomial-time with Shor's algorithm [18] on a sufficiently powerful quantum computer. Moreover, using Grover's algorithm [19] by quantum computers reduces the effective key length by a factor of two by providing quadratic speed up for the unstructured search, which applies to searching the key in symmetric schemes like AES and 3DES.

Table 1 presents a list of acronyms that are used frequently in this review article.

The breakability of existing classical cryptosystems by quantum computers is a potential threat to the present and a more serious threat to future information security systems [20–22]. Presently, an eavesdropper can intercept cryptograms while waiting for their decryption once a sufficiently large quantum computer is technologically available. This is because present-day cryptographic algorithms rely on hard mathematical problems, which are solvable on quantum computers. Therefore, researchers are exploring quantum-resistant cryptographic algorithms. One solution is the use of PQC algorithms

Table 1 List of key acronyms

Label	Explanation
AES	Advanced encryption standard
AI	Artificial intelligence
BLISS	Bimodal lattice signature scheme
BIKE	Bit flipping key encapsulation
CBC	Code-based cryptography
CCA	Chosen-ciphertext attack
CISA	Cybersecurity and Infrastructure Security Agency
CE	Consumer electronics
CPS	Cyber-physical systems
DES	Data encryption standard
DH	Diffie–Hellman
DLP	Discrete logarithm problem
ECDSA	Elliptic curve digital signature algorithm
HFE	Hidden field equations
HILA5	High-interest low-complexity algorithm
HQC	Hamming quasi-cyclic
IF	Integer factorization
IoT	Internet of things
LBC	Lattice-based cryptography
MAC	Message authentication code
NIST	National Institute of Standards and Technology
NSA	National Security Agency
PQC	Post quantum cryptography
QKD	Quantum key distribution
QML	Quantum machine learning
QoE	Quality of experience
QROM	Quantum random oracle model
QC-MDPC	Quasi-cyclic moderate density parity-check
RSA	Rivest, Shamir, Adleman
SHA	Secure hash algorithm
SIDH	Supersingular isogeny Diffie–Hellman
SIKE	Supersingular isogeny key encapsulation
SIS	Short integer solution
XMSS	eXtended merkle signature scheme

[23–28], which can resist adversarial attacks from classical as well as quantum computers. PQC seeks to propose cryptographic primitives that remain secure even in the presence of quantum adversaries [29, 30], ensuring the long-term security of information exchange and storage as classical cryptography relies on the hardness of problems such as factoring large integers or computing discrete logarithms [31]. PQC leverages alternative mathematical assumptions to achieve security. Lattice-based cryptography (LBC) [32], code-based cryptography (CBC) [33, 34], hash-based cryptography (HBC) [35], multivariate polynomial cryptography (MPC) [36], isogeny-based cryptography (IBC) [37], non-commutative cryptography (NCC) [38] and other emerging paradigms represent promising avenues for constructing PQC primitives.

In addition to developing new cryptographic primitives, the standardization and evaluation of PQC algorithms are essential for their widespread adoption and integration into real-world systems. Organizations like the national institute of standards and technology (NIST) [39], the national security agency (NSA), and the cybersecurity and infrastructure security agency (CISA) have launched initiatives to solicit and evaluate candidate PQC algorithms through open competitions and peer-reviewed processes. However, adopting the PQC suite of algorithms will require a transition period of several decades, which will consider various factors, including compliance, algorithm performance, security and ease of implementation. These efforts aim to guarantee the availability of secure cryptographic solutions that can preserve sensitive information and enable secure communications in light of the forthcoming advancements in quantum computing technology.

1.1 Existing literature and their limitations

In the last couple of years, a few reviews, tutorials, and survey articles covering PQC algorithms have been published [24, 26, 40–64]. Table 2 compare the most recent, relevant, and popular articles on this topic based on their main contributions. More specifically, Chamola et al. [50] addressed the applications of quantum computers, threats to classical cryptographic systems, and the future of information security in the post-quantum era. The authors believe that QKD uses quantum mechanical phenomena that can combat the vulnerabilities of quantum computers. Similarly, Balamurugan et al. [51] explored research directions in the field of PQC and, precisely, CBC. In this study, the authors also discussed the NIST's standardization efforts. The authors believe that CBC's scope is to be accepted as a complete cryptosystem similar to its counterpart's methods.

Yalamuria et al. [52] carried out a systematic literature review study in which the authors outlined six categories of PQC, namely, CBC, HBC, LBC, symmetric-key-based cryptosystems, MPC and IBC. The authors of this study concluded that the LBC PQC method is the most popular among the others in the existing literature. The challenges, such as hardware and software implementation of PQC families, are also highlighted, which the authors believe require future research. Joseph et al. [26] discussed PQC transition timings and different strategies to secure systems against various types of quantum attacks. To reduce transition risks, the authors of the paper [26] also investigated methods for integrating PQC with pre-quantum cryptography. In addition to offering various suggestions to assist organizations in achieving a swift and seamless PQC transition, the authors proposed standards to begin experimenting with immediately. Recent developments in PQC algorithms with an emphasis on networking and device issues

Table 2 Comparison with related reviews, tutorials, and survey articles on PQC

Year	Literature	Consumer electronics	PQC	Classical crypt	Implementation	Complexity	Standardization	Comparison	Research gaps
2024	Ref. [64]	×	×	∂	×	×	×	✓	✓
	Ref. [62]	×	✓	∂	×	×	×	✓	✓
	Ref. [61]	×	✓	×	×	×	×	✓	✓
2023	Ref. [59]	×	✓	✓	×	×	✓	∂	✓
	Ref. [58]	×	✓	✓	×	∂	×	✓	✓
	Ref. [57]	×	✓	✓	×	∂	×	✓	✓
	Ref. [56]	×	✓	✓	×	∂	×	✓	✓
	Ref. [54]	×	×	×	×	∂	×	∂	✓
2022	Ref. [53]	×	×	×	✓	×	✓	✓	✓
	Ref. [26]	×	✓	×	✓	×	✓	✓	✓
	Ref. [52]	×	×	×	✓	×	×	∂	✓
2021	Ref. [51]	×	✓	×	×	×	✓	✓	✓
	Ref. [50]	×	✓	×	×	×	∂	✓	✓
Our review		✓	✓	✓	✓	✓	✓	✓	✓

(✓), (✗) and (∂) show that the topic has been covered, not covered, and only a portion of the topic has been covered, respectively

were summarized by Zeydan et al. [53]. The authors initially provided an overview of the PQC development process's life-cycle components, followed by quantum-resistant network platforms and, lastly, the most recent developments in standardization and commercial/open-source frameworks. Finally, the authors addressed the challenges and potential research directions for developing a PQC-based networking system for the telecommunications industry.

Bavdekar et al. [54] analyzed the vulnerabilities of the classical cryptosystems raised from intruders using quantum computers. The authors of the study [54] also discussed various PQC types, NIST's PQC standardization efforts, and the status of ongoing projects, as well as performance comparison of PQC algorithms on different platforms. The authors concluded their study with future research directions in the field of PQC. In another work, Subramani et al. [56] covered classical and quantum cryptographic methods and presented a comparison with respect to time, security level, and data classification. Similarly, Shaller et al. [57] provided a comprehensive review of different types of attacks and countermeasures in PQC to portray a roadmap of PQC standardization led by NIST. More specifically, the authors of the study [57] covered side-channel attacks against PQC schemes and their countermeasures in major PQC schemes, i.e., the final NIST candidates.

In another review article, Dam et al. [58] discussed public cryptographic methods and digital signature schemes. In addition, the authors explored the NIST standardization efforts along with the competition regarding selecting the best candidate for the expected standard. The article [58] summarized recent literature on PQC, thereby showing that PQC still has much potential and is a promising avenue for the research process of cryptographic encryption. Iqbal et al. [59] outlined the numerous study avenues researched in PQC and, more particularly, the various CBC research aspects. An essential contribution of this research is the identification of unexplored prospective research avenues in CBC research from the viewpoint of codes. In addition, the authors

investigated the applicability of these algorithms to IoT devices and the possible future aspects and challenges. Xie et al. [61] provided a thorough tutorial on hardware circuit and systems design for PQC, describing fundamental arithmetic operations and algorithm-specific features for various PQCs. Gharavi et al. [62] presented a review article to classify types of PQC and their standard primitives and study whether they can provide security for blockchain-based IoT applications. Lastly, Khan et al. [64], in their recent work, offered a review article about PQC for UAV networks. The authors first looked at the security and privacy landscape of UAVs, including threats and requirements, and then discussed PQC and its families.

1.2 Contributions and article organization

While many studies have discussed different aspects of PQC in different domains, there is hardly an article that has focused on CE or the next-generation CE. With the increasing threat posed by quantum computers, there is an urgent need to secure existing systems, making this the perfect time to present a clear, detailed, and up-to-date review of the PQC. Although significant progress has been made in developing and testing PQC algorithms, ongoing research means the threats remain unresolved and require continuous attention. To our knowledge, this is the first review article that focuses entirely on PQC on CE or next-generation CE. The main contributions of this work include:

- This article provides an overview of next-generation CE, focusing on their rapidly evolving security challenges in the presence of quantum computers.
- A major focus of the study is PQC, which is presented as a vital solution for securing next-generation CE. The article details the advantages of PQC, particularly its ability to address the vulnerabilities that quantum computers pose to classical cryptographic systems.
- The practical implementation of PQC in next-generation CE is also discussed in this work.
- Finally, this paper emphasizes the key challenges and outlines future directions, providing a foundation for innovative advancements in the development of secure next-generation CE systems.

The article's organization is as follows: Section 2 discusses next-generation CE and key challenges to security. Section 3 covers PQC and its types. Section 4 covers the implementation of PQC in the next-generation CE. Section 5 is dedicated to challenges and future directions. Section 6 is devoted to the conclusion. The illustration of the article's organization is provided in Fig. 2.

1.3 Review methodology

In this study, we used a fundamental procedure [65] to identify and filter the existing literature in the fields of PQC and CE. First, we searched through digital databases, specifically the well-known ACM Digital Library and IEEE Xplore. They are notable in computer science and related technical fields under the umbrella of ACM and IEEE. We also explored the most recent research through the Cryptology ePrint Archive; however, the majority of the papers on this database are not subject to peer review. Since the archive has a well-known reputation for processing the most recent material quickly, we were unable to avoid it. Similarly, this fits our aim of reviewing the most recent information on

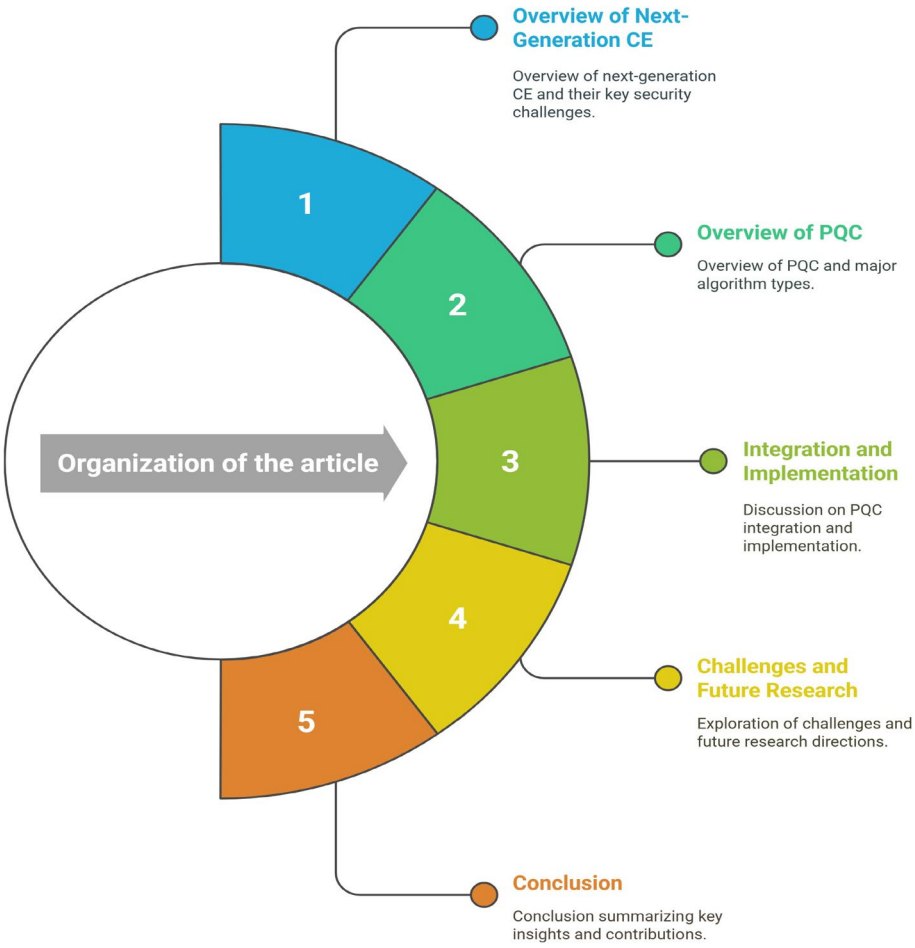

Fig. 2 Overview of the article's structural organization

PQC through the articles that we found are thoroughly evaluated. In relation to industrial cryptography, using one of the author's (G. Vidal) knowledge, we defined an initial code of relevant literature on this topic. Then, we supplemented such literature through IEEE Xplore and ACM databases and by obtaining references from such an initial set of papers. Also, PQC is a fast-moving field, and therefore, blogs related to PQC were read frequently when writing this article in order to track developments in this area.

In our scope, the main criteria for selecting a relevant article for this review are as follows:

- Be on the topic covering PQC in general, PQC in consumer electronics, implementations and standardizations.
- Be published in 2021 and after, and their comparisons are shown in Table 2.

We used the keywords "Post-quantum cryptography," "Quantum-resistance cryptography," "Consumer electronics," "Post-quantum cryptography in consumer electronics," "PQC," " Cybersecurity issues in consumer electronics," "Quantum cryptography types," "Quantum attacks on classical cryptography," and similar keywords to identify possibly relevant publications.

During the initial search, 1,610 articles were identified. After careful filtering, 100 publications were selected for detailed review and categorization. Among these, 50

publications were relevant to the scope of the review, including 13 specifically focused on the comparisons, as shown in Table 2, which is further illustrated in Fig. 3.

2 Next-generation consumer electronics

The CE landscape is undergoing an exponential shift driven by technological advancements, increasing user demand for seamless experiences. The next-generation CE is marked by intelligent, interconnected devices designed to enhance convenience in everyday life. These devices encompass wearable technologies, smart home solutions, augmented and virtual reality systems, advanced robotics, and much more. The integration of cutting-edge technologies like edge computing, the IoT, Blockchain [66, 67] and AI/ML is making CE more intelligent and highly responsive than ever before. Besides, 5G offers cloud computing functionalities [68], when integrated into a CE environment, can relieve resource-constrained CE devices from processing heavy computational tasks. The basic concept of the edge computing framework to deliver on-demand computing services to CE devices through e-URLLC wireless connectivity was proposed by Nawaz et al. [69]. Within this framework, the latency caused by offloading computing tasks from CE devices to an edge computing platform is expected to be nearly equivalent to performing those tasks locally. This is made possible by advancements in link speed, reliability, and low-latency performance characteristic of the Beyond 5G (B5G) era. The innovations associated with this approach include: (1) Eliminating cabling requirements such as those between a CPU and monitor, streamlining device designs, and enhancing

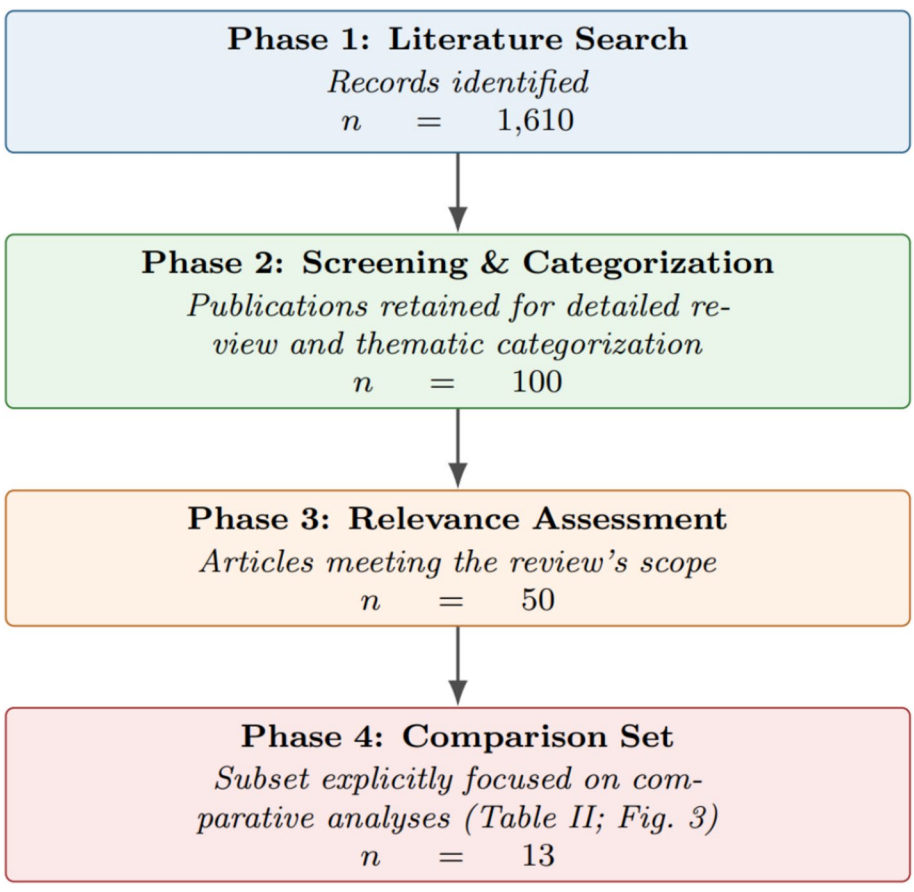

Fig. 3 Research methodology flow

portability. (2) Inter-chip communication via THz line-of-sight links, which enables high-speed, wireless communication between components within CE circuitry for faster and more efficient data transfer. (3) Processor-less (or minimal-processor) devices to reduce onboard processing power, relying on edge platforms for computational tasks, resulting in more straightforward and potentially more cost-effective designs. These advancements open up exciting possibilities for next-generation CE devices, offering greater flexibility, efficiency, and innovation in their design and functionality [69]. Fig. 4 depicts a sample architecture of next-generation CE devices using an edge computing platform On the one hand, edge computing technology is emerging as a cornerstone of next-generation CE, enabling smarter, faster, and more reliable systems that align with the needs of modern consumers. On the other hand, this progress comes with increased cybersecurity challenges. Leveraging edge or cloud computing introduces potential vulnerabilities, as data exchanged between consumer devices and computing platforms can be exposed to various cyber threats, including message interception, unauthorized access, availability attacks, spoofing, and more [70, 71]. As the landscape of cyber threats evolves with new trends, conventional cybersecurity measures designed for traditional CE may no longer be sufficient. Due to the presence of quantum computers, next-generation CE devices will face more sophisticated threats. Therefore, to address these challenges, robust and innovative security strategies must be developed to protect the integrity and functionality of these advanced systems.

2.1 Key challenges to security of next-generation CE

As we know, in today's world, users typically demand advanced features and functionalities in CE devices, and this demand continues to grow. Manufacturers face the formidable challenge of balancing cost, features, and security. It is not easy to meet these

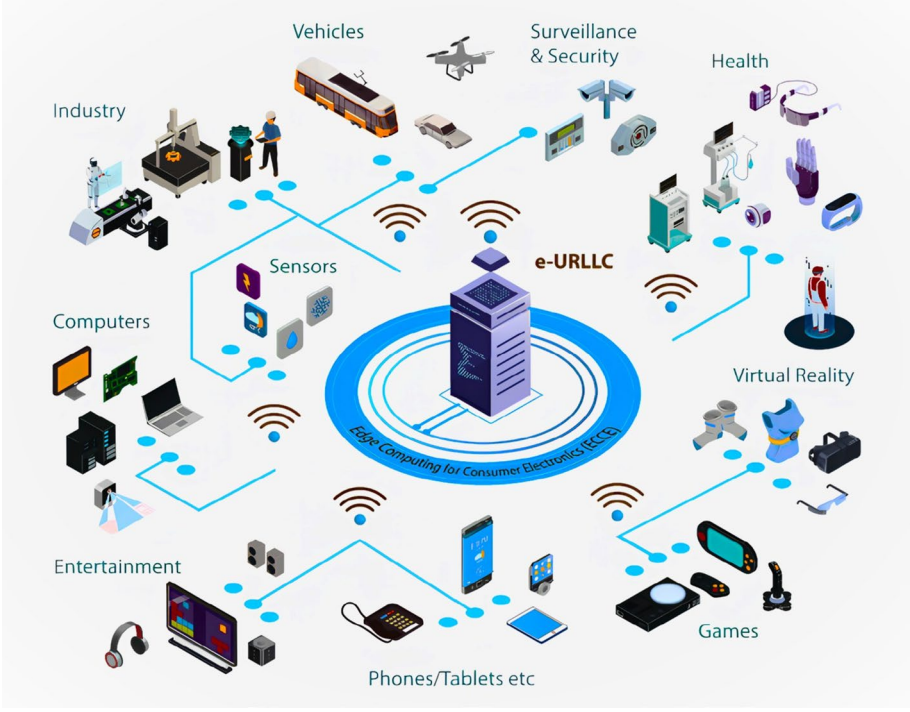

Fig. 4 A sample architecture of next-generation CE devices using edge computing platform [69]

expectations while ensuring devices remain affordable, functional, and secure. In this section, we explore the key challenges that manufacturers may encounter while integrating security and privacy measures in these devices, as illustrated in Fig. 5. These hurdles highlight the complexities of delivering cutting-edge CE that is not only innovative and cost-effective but also resilient against evolving cybersecurity threats. .

- *Limited Resources:* Most CE devices are poorly designed with security considerations in mind. They are generally designed to fit into users' budgets, and most of the time, these devices have inadequate processing and storage capabilities.
- *Mobility:* At least 80 percent of CE devices are movable using wireless communication technologies; therefore, it is difficult to maintain essential security characteristics such as authentication and authorization for all individual devices in a dynamic environment based on their location.
- *Wireless connectivity:* Next-generation CE devices rely on ubiquitous open wireless channels, making them prone to cybersecurity breaches.
- *Real-Time Services:* Most of the CE devices are designed to meet the real-time response requirement of users, which incurs a high latency phase because of the extensive network traffic in the path from the user to the server.

The challenges facing next-generation CE underscore the importance of enhancing security and privacy measures to preserve key attributes, including confidentiality, integrity, availability, authentication, trust, and data privacy protection. The advent of 5G and B5G mobile networks exposes these devices to more advanced intentional and unintentional attacks [72]. Similarly, with the presence of quantum computing technology, the urgency of addressing security vulnerabilities has received more attention. Likewise, traditional cryptographic methods, including both symmetric and asymmetric methods, are already vulnerable to being broken by the immense computational power of quantum computers, rendering them unsuitable for future-proof security. This is where quantum-resistant security schemes become essential, which can be provided by PQC. PQC algorithms are designed to resist both classical and quantum-based attacks while remaining compatible with current communication and processing systems. By integrating PQC

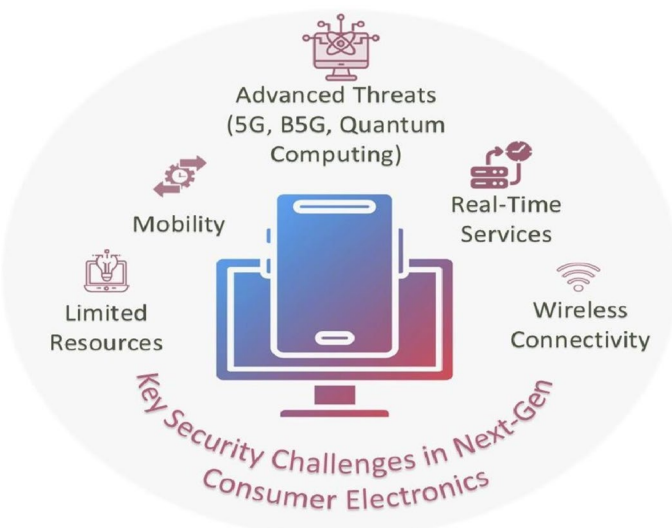

Fig. 5 Key security challenges in next-generation CE

into next-generation CE devices, we can ensure their security remains robust even against emerging quantum threats. This shift will help protect sensitive data, maintain user trust, and safeguard the critical functionality of these devices in a highly connected world. Adopting PQC isn't just an upgrade; it's a necessity for building a secure future in CE.

The principles of quantum mechanics have a strong association with the postulates of Euclidean geometry [73], which are applied to the CE devices constructed from elementary particles. The fundamental concepts of quantum entanglement, quantum superposition, and the no-cloning theorem provide a great deal of parallelism in computing principles, and results have also been recently shown to support this. Both binary states can be held simultaneously by a qubit. As a result, these computer systems are capable of producing statistical data patterns that defy logic, which is not possible with traditional computing systems. In recognition of these potentials, new disciplines like "Quantum Computer Science" and "Quantum Information Science" have also originated.

3 Post-quantum cryptography

The development of the first universal quantum computing model by David Deutsch, based on physical principles and the Church-Turing hypothesis [74], provided the theoretical basis for evaluating the security of PQC primitives. Figure 6 depicts advances in cryptography and quantum processing. At its core, quantum computing operates on quantum bits or qubits [75], which, unlike classical bits, exist in multiple states such as both 0 and 1 at the same time due to the principle of superposition [76–80]. This property enables quantum computers to perform computations on multiple possible inputs simultaneously, leading to an exponential increase in computation power [81]. Shor's algorithm [18], developed by mathematician Peter Shor in 1994, stands as one of the most prominent examples of the computation power of quantum computing. Shor's algorithm efficiently factors large integers and solves the discrete logarithm problem, a foundational step in many asymmetric cryptographic schemes, such as RSA and ECC [35, 41, 82]. Similarly, Grover's algorithm [19], proposed by Lov Grover in 1996, demonstrates another facet of quantum computing's impact on classical cryptography in

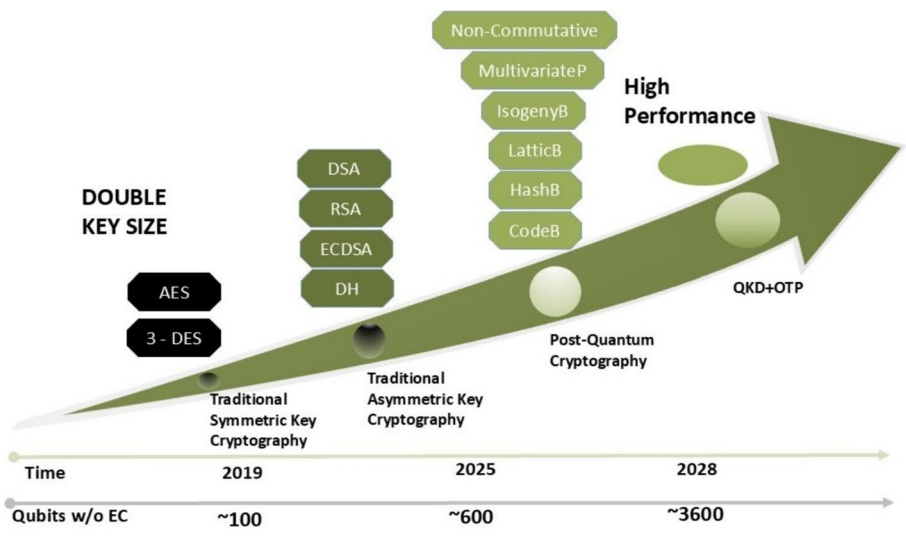

Fig. 6 Advances in cryptography and quantum processing [88]

Table 3 Comparison of the widely adopted symmetric cryptosystems and their security levels against pre- and post-quantum attacks

Cryptographic algorithm	Type	Function	Pre-quantum security level (in bits)	Post-quantum security level (in bits)	Status
AES-128 [99]	Block cipher	Encryption	128	64	Cracked by Grover's algorithm
AES-256 [99]	Block cipher	Encryption	256	128	Cracked by Grover's algorithm
DES [100]	Block cipher	Encryption	56	28	Cracked by Grover's algorithm
Salsa20 [101]	Stream cipher	Encryption	256	128	Cracked by Grover's algorithm
GMAC [102]	MAC	Authentication	128	128	No impact
Poly1305 [103]	MAC	Authentication	128	128	No impact
SHA-256 [104]	Hash function	Hashing	256	128	Cracked by Grover's algorithm
SHA-3 [105]	Hash function	Hashing	256	128	Cracked by Grover's algorithm

Table 4 Comparison of the widely adopted asymmetric cryptosystems and their security levels against pre- and post-quantum attacks

Cryptographic algorithm	Type	Function	Pre-quantum security level	Post-quantum security level	Status
RSA-3072 [106]	Asymmetric encryption	Encryption	128 bits	Broken	Cracked by Shor's algorithm
RSA-3072 [106]	Asymmetric digital signature	Digital signatures	128 bits	Broken	Cracked by Shor's algorithm
DH-3072 [107]	Asymmetric key exchange	Key exchange	128 bits	Broken	Cracked by Shor's algorithm
DSA-3072 [108]	Asymmetric digital signature	Digital signatures	128 bits	Broken	Cracked by Shor's algorithm
256-bit ECDH [109]	Asymmetric key exchange	Key exchange	128 bits	Broken	Cracked by Shor's algorithm
256-bit ECDSA [110]	Asymmetric digital signature	Digital signatures	128 bits	Broken	Cracked by Shor's algorithm

symmetric schemes, such as AES and 3DES [83–85]. Grover's algorithm provides a quadratic speedup over classical algorithms for searching an unsorted database and reduces the security strength of symmetric encryption and hash functions by halving their effective key lengths [86, 87].

Tables 3 and 4 compare the widely adopted symmetric and asymmetric cryptosystems, along with their security levels, against both pre- and post-quantum attacks. These tables reflect the impact of Shor's and Grover's algorithms on classical cryptosystems, giving the impression that quantum computers destroy the viability of asymmetric cryptography, leaving only symmetric cryptography alive; however, with the option of considering larger key sizes. Asymmetric cryptosystems, including the RSA, DSA, DH, ECDH key exchange, and the ECDSA, are insecure against quantum computing attacks, which are cracked by Shor's algorithm. The security toughness of the RSA cryptographic method generally relies on the hardness of factoring large bi-prime numbers, also known as the integer factorization (IF) problem [50, 52, 89, 90]. In contrast, ECC acknowledges that ECDL problems are challenging to solve [91–93]. However, both IF and the DLP are

considered to be difficult for classical computers to solve [94]. Still, they can be cracked in polynomial time by a quantum computer large enough to run Shor's algorithm. In today's time, however, small experimental quantum computers are not equipped enough to solve practical ciphers, and the researchers are estimating the quantum resources needed to achieve such a goal in the near future [95]. Likewise, by cracking the symmetric cryptographic methods, Grover's algorithm can potentially reduce the security strength of existing ciphers by offering a quadratic speed-up for exhaustive key searching. The attack's practical implications are still debated, though, because Grover's algorithm necessitates running queries sequentially [96]. Nevertheless, researchers have calculated that Grover's search against the AES with 128, 192, and 256-bit security can be performed by a quantum architecture with 2,953, 4,449, and 6,681 qubits, respectively [97]. Consequently, it is well accepted that quantum computers seriously threaten both symmetric and asymmetric cryptosystems; however, the danger to asymmetric cryptography is far greater than that to symmetric cryptography [24, 98]. Security level reductions attributed to Grover's algorithm are theoretical estimates based on the assumption of an ideal quantum computer equipped with oracle access to the cryptographic primitive. Figure 7 presents an Illustration of encryption and decryption mechanisms in symmetric and asymmetric cryptography.

Shor's Algorithm Example:

Shor's algorithm efficiently factors large composite integers by transforming the problem into one of period finding. As an illustrative case, consider factoring $N = 15$. We select a random integer $a < N$ such that $\gcd(a, N) = 1$; for example, let $a = 2$. Define the function:

$$f(x) = a^x \mod N = 2^x \mod 15$$

This function is periodic, and the quantum subroutine (based on the Quantum Fourier Transform) is used to find its period r, such that:

$$f(x + r) = f(x)$$

In this case, the period is found to be $r = 4$. Once the period is known, non-trivial factors of N can be obtained via:

$$\gcd(a^{r/2} \pm 1, N)$$

Substituting the known values:

$$\gcd(2^2 - 1, 15) = \gcd(3, 15) = 3, \quad \gcd(2^2 + 1, 15) = \gcd(5, 15) = 5$$

Thus, the prime factors of 15 are 3 and 5. While this example is trivial, Shor's algorithm generalizes to efficiently factor large integers, thereby compromising the security of RSA, ECC, and other public-key systems relying on the hardness of integer factorization or discrete logarithms.

Grover's Algorithm Example:

Grover's algorithm offers a quadratic speedup for unstructured search problems. In cryptographic terms, this affects brute-force key search. Suppose a symmetric cipher uses a key of size n, yielding a search space of size $N = 2^n$. Classically, the expected

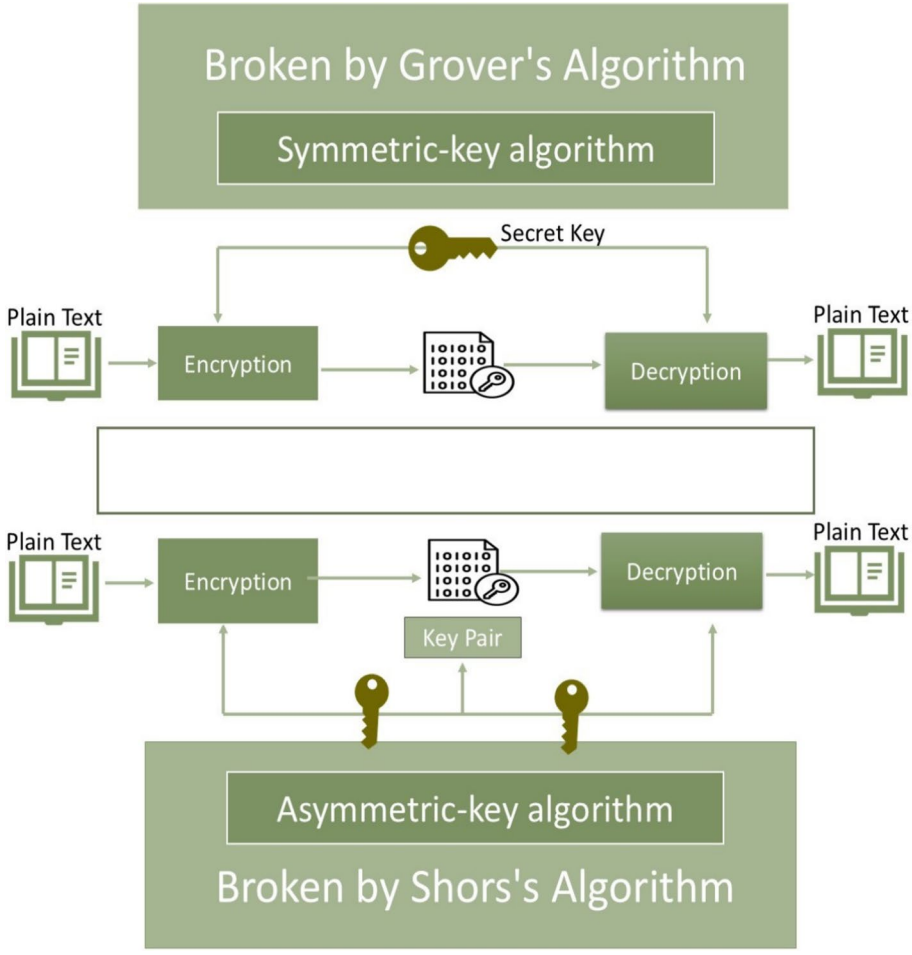

Fig. 7 Illustration of encryption and decryption mechanisms in symmetric and asymmetric cryptography

number of operations to find the correct key is $O(2^n)$. With Grover's algorithm, this is reduced to $O(2^{n/2})$.

For instance, a cipher with a 128-bit key requires:

$$O(2^{128}) \text{ operations classically}, \quad O(2^{64}) \text{ operations quantumly}$$

This effectively halves the brute-force security of symmetric ciphers. To maintain the same security level in a post-quantum setting, key sizes must be doubled. As a result, AES-256 is generally recommended to ensure quantum-resilient symmetric encryption.

PQC operates under the assumption that potential adversaries are equipped with advanced quantum computers and necessitate cryptographic methods resilient to quantum attacks [111]. The primary objective of PQC is to uphold cryptographic functionality and adaptability by developing algorithms and protocols that can withstand quantum threats [112]. This requires that classical cryptography stop relying only on the assumed hardness of IF and the DLP and, therefore, start using problems that even quantum computers could not efficiently solve [94]. Despite the fact that powerful quantum computers are still unavailable commonly, there have been requests for the Internet to get ready for the transition to PQC [49, 113, 114]. There are two reasons for this: first, pre-quantum public-key cryptosystems have been incorporated into many applications and protocols, and second, secret communications can be stored and decrypted with QC later [115].

As per the guidelines, a variety of PQC solutions have been developed to meet the PQC needs and criteria. Each of the proposed algorithms falls into one of the PQC families based on its mathematical foundation. CBC [116], HBC [117], MBC [118], IBC [119], NCC [120], and LBC [121] are the main families.

LBC holds excellent promise for PQC, used to construct cryptographic primitives that involve lattices [122–124], either in the construction itself or enjoy strong security proofs based on worst-case hardness [44, 125–128]. Lattice-based constructions are considered secure, assuming that specific well-studied computational lattice problems cannot be solved efficiently on even quantum computers [129]. CBC represents a critical approach within PQC, designed to safeguard data using error-correcting codes against potential quantum computing threats. Unlike classical cryptographic methods such as RSA and ECC, which rely on number-theoretic problems vulnerable to quantum attacks, CBC utilizes the computationally intensive NP-hard problem of decoding arbitrary linear codes [34, 116, 130]. HBC [131, 132] relies on hash functions to achieve various essential security goals, such as data integrity, authentication, and digital signatures. A hash function is a mathematical algorithm that transforms an input (or 'message') into a fixed-size string of bytes, typically a hash value (or 'digest'). The output usually is a much smaller, fixed size, regardless of the input size. HBC has a rich history dating back to the 1970s, beginning with the formalization of cryptographic hash functions [133]. Ralph Merkle's introduction of Merkle trees and the Merkle-Damga˚rd construction laid the foundation for hash-based cryptography, while Leslie Lamport's and Merkle's subsequent developments, including the Lamport signature scheme and Merkle Signature Scheme (MSS) [134–137], paved the way for more efficient and secure hash-based digital signatures. MPC refers to a class of public-key cryptosystems based on multivariate polynomial equations over finite fields. IBC is founded on mathematical structures known as isogenies, which are special functions between elliptic curves that preserve their group structures. In this context, an isogeny graph is a structure where nodes represent isomorphism classes, and edges represent the isogenies between curves. NCC is a special type of PQC that explores cryptographic schemes based on mathematical structures where the order of operations affects the outcome. Dissimilar to classical cryptographic systems that often rely on commutative operations, such as the multiplication of numbers (where a.b = b.a), NCC utilizes operations that do not necessarily commute, such as matrix multiplication or operations in non-abelian groups.

3.1 Mathematical Basis of PQC Primitives

- **Code-Based Cryptography (CBC)** relies on the hardness of decoding random linear codes. Given a public generator matrix $G \quad \mathbf{F}^{k\times n}$, the cryptographic challenge is to decode a received word $\mathbf{c} = \mathbf{m}G + \mathbf{e}$, where $\mathbf{m}$ is the message and $\mathbf{e}$ is a sparse error vector with Hamming weight t. The decoding problem is to recover $\mathbf{m}$ given $\mathbf{c}$, which is known to be NP-hard.
- **Hash-Based Cryptography (HBC)** constructs signatures using cryptographic hash functions H. The security relies on the preimage and collision resistance of H. For example, a one-time signature can be constructed by publishing a hash chain

$$h_0 = \text{seed}, \quad h_i = H(h_{i-1}), \quad \text{for } i = 1, \ldots, n.$$

Verification involves checking the hash path against the public value.

- **Multivariate-Based Cryptography (MBC)** depends on the difficulty of solving systems of multivariate quadratic equations over finite fields. Given a public system

$$\mathbf{P}(\mathbf{x}) = (P_1(\mathbf{x}), ..., P_m(\mathbf{x})) = \mathbf{y},$$

where each P_i is a quadratic polynomial and **y** is the output, the problem is to find **x** satisfying the system, which is known to be NP-hard.

- **Isogeny-Based Cryptography (IBC)** utilizes the hardness of computing isogenies between supersingular elliptic curves. Given two elliptic curves E_1 and E_2, the problem is to find an isogeny ϕ: $E_1 \rightarrow E_2$ of a given degree, which is believed to be difficult even for quantum computers.
- **Non-Commutative Cryptography (NCC)** relies on hard problems in non-commutative algebraic structures such as braid groups or matrix groups. For example, given elements a, b in a non-commutative group G, the conjugacy search problem is to find x such that

$$b = x^{-1}ax,$$

which is computationally hard.

- **Lattice-Based Cryptography (LBC)** is based on the hardness of lattice problems such as the Learning With Errors (LWE) problem. Given a matrix $A \in \mathbf{Z}^{m \times n}$, secret vector $\mathbf{s} \in Z^n$, and error vector $\mathbf{e} \in Z^m$, the LWE samples are

$$\mathbf{b} = A\mathbf{s} + \mathbf{e}(\mathrm{mod} q).$$

Recovering **s** from (A, **b**) is conjectured to be hard.
These mathematical foundations underlie the security assumptions of PQC schemes implemented on resource-constrained CE devices. Later sections explore the practical implementation considerations and performance benchmarks of these schemes.

4 Implementation of PQC in next-generation CE

With the ubiquitous connectivity of 5G networks, next-generation CE devices will become increasingly vulnerable to advanced cyber threats, including those posed by the rise of quantum computers. Figure 8 presents a PQC framework securing CE devices, authentication, data transmission, and cloud protection against quantum threats.

Implementing PQC in next-generation CE, however, requires a careful strategy that addresses technical, operational, and security challenges while maintaining performance and user experience. Figure 9 highlights the key components of PQC integration in next-generation CE, showcasing the role of hardware, software, communication protocols, standardization, and edge computing in ensuring quantum-resistant security.

4.1 Hardware implementation

- *Hardware-Optimized Cryptographic Modules:* Hardware-optimized cryptographic modules for PQC are revolutionizing the security landscape for next-generation CE, like smartphones, IoT gadgets, and wearables. Designed to handle the demanding

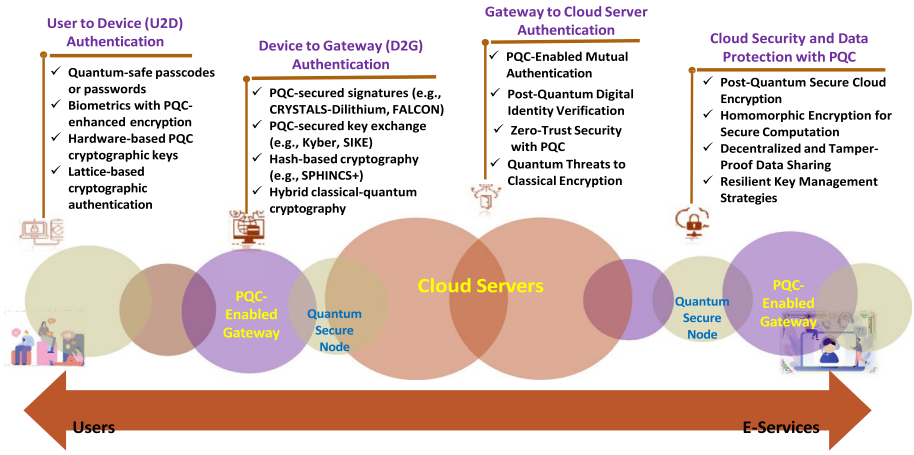

Fig. 8 Post-quantum cryptography framework securing CE, authentication, data transmission, and cloud protection against quantum threats

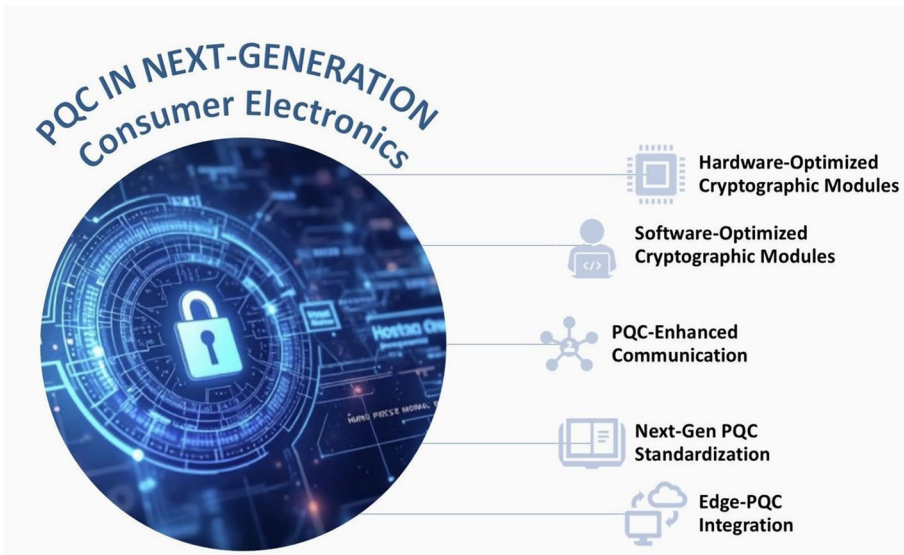

Fig. 9 Overview of PQC integration in next-generation CE, highlighting key components for enhanced security

computational requirements of PQC algorithms such as Kyber, Dilithium, and Falcon, these modules ensure fast processing while conserving battery life, an essential feature for portable electronics. They also include advanced safeguards against sophisticated attacks, such as power analysis and fault injection, making them exceptionally secure. By utilizing specialized hardware, such as ASICs or FPGAs, these modules outperform traditional software-only approaches in terms of speed and energy efficiency, enabling real-time encryption even in devices with limited resources.

- *Secure Elements:* The integration of PQC into secure hardware elements, such as trusted platform modules (TPMs) and hardware security modules (HSMs), provides a robust layer of protection for cryptographic operations performed on CE devices. These modules create a physically secure environment, shielding key processes like encryption and signature generation from physical tampering and side-channel attacks on CE devices. Additionally, low-power processors, such as

the ARM Cortex-M series, can be adapted to incorporate PQC-enabled instruction sets, bringing quantum-resistant security to energy-efficient CE devices. An implementation example of PQC on ARM Cortex-M series involves integrating the CRYSTALS-Dilithium digital signature scheme into a secure boot process on an STM32F407 (Cortex-M4) microcontroller [138]. Using the optimized pqm4 library, the device verifies firmware authenticity by checking a Dilithium signature at startup. The public key is stored in flash, and the firmware image is verified using SHAKE256 and the dilithium verify() function. On Cortex-M4, Dilithium-II verification takes 2.8 ms with a signature size of 2.4 KB, making it feasible for resource-constrained CE devices. This adaptation demonstrates that Cortex-M devices can achieve quantum-resistant security while maintaining efficiency and reliability.

- *IoT Device-Specific Design*: IoT-specific hardware for PQC is designed to meet the unique challenges of small, battery-powered devices. These lightweight solutions optimize performance while minimizing power and resource usage, enabling secure encryption and key management within the limited capabilities of IoT gadgets. By leveraging efficient architectures and power-aware designs, these systems make quantum-resistant security practical for devices like smart sensors and wearables. This ensures strong data protection and secure communication across the growing Internet of Things ecosystem.

4.2 Software implementation

- *Cryptographic Libraries:* Incorporating PQC algorithms into established libraries, like open quantum safe (OQS), makes adopting quantum-resistant security straightforward and accessible. OQS provides open-source, PQC-ready libraries that are compatible with widely used protocols, such as TLS and SSH, allowing for seamless integration with existing systems. These libraries also support hybrid modes, enabling the use of both classical and PQC algorithms during the transition to complete quantum-safe solutions. Regular updates ensure alignment with evolving PQC standards, providing developers with a reliable, flexible, and future-proof approach to implementing quantum-resistant security in real-world applications.
- *Device Firmware Updates*: Deploying PQC through over-the-air (OTA) updates allows devices to securely receive firmware upgrades, ensuring they stay protected against emerging threats. This seamless integration of PQC ensures devices are kept up to date with minimal user effort. Additionally, implementing secure boot chains with PQC-based signature verification ensures that devices verify the integrity of their firmware at startup, preventing tampering and malicious code from compromising the system. Together, OTA updates and secure boot chains provide robust, ongoing security for IoT and CE. In addition, PQC enhances firmware authenticity and integrity by replacing classical digital signature schemes (such as RSA or ECDSA) with quantum-resistant algorithms like CRYSTALS-Dilithium or SPHINCS + . These schemes ensure that even with the advent of quantum computers capable of breaking traditional cryptography, the bootloader can still securely verify the firmware before it is executed. This adaptation is especially critical for next-generation CE, where long-term security and trusted firmware updates are essential.
- *OS level Support:* OS-level support for PQC in real-time operating systems (RTOS) like FreeRTOS or Zephyr involves updating the core software to incorporate

quantum-resistant cryptographic algorithms. By implementing PQC algorithms, such as Kyber for key exchange or Dilithium for authentication, these RTOS can ensure secure communication and data protection in embedded systems, even in the era of quantum computing.

4.3 Communication protocols

- *Upgrading Network Protocols*: Upgrading network protocols like MQTT, HTTP/2, or CoAP to incorporate PQC enhances session establishment and message encryption, providing stronger security against potential quantum threats. For instance, integrating PQC algorithms into TLS 1.3 can secure transmission in CE devices by enabling quantum-resistant encryption during data exchanges. By replacing or supplementing classical algorithms with PQC schemes, such as Kyber for key exchange or Falcon for digital signatures, these protocols ensure long-term security, protecting sensitive communications even as quantum computing advances. This upgrade is essential for future-proofing IoT networks and ensuring robust encryption in emerging technologies.

  Integrating PQC into communication protocols such as TLS 1.3 introduces measurable overhead in both latency and bandwidth. This is mainly due to the increased computational complexity and larger key and signature sizes associated with quantum-resistant algorithms, such as Kyber and Dilithium. On embedded platforms such as the ARM Cortex-M4, PQC-enabled TLS handshakes have been shown to increase latency by tens to hundreds of milliseconds and expand bandwidth usage by several kilobytes per session. In cloud-based scenarios, experimental evaluations of hybrid TLS handshakes (e.g., ECDHE + Kyber) report an additional 0.25 ms of latency on the client side and approximately 2.3 KB of increased handshake data [139]. While these overheads pose challenges, particularly for resource-constrained or real-time applications, they are generally considered acceptable for many next-generation CE systems, especially when optimized or hybrid implementations are used.
- *Authentication Mechanisms:*Adopting PQC-based digital signatures, such as CRYSTALS-Dilithium and Falcon, for device authentication significantly enhances security in environments like smart home systems. These algorithms provide quantum-resistant solutions for signing critical data, such as firmware updates or session keys, ensuring that only authorized devices can authenticate and securely exchange information. By leveraging PQC digital signatures for key operations, smart home systems are protected against future quantum computing threats, ensuring the integrity and authenticity of communications and device interactions in increasingly connected environments. This strengthens overall trust and security across IoT ecosystems.
- *Quantum-Resistant Key Exchange:*Implementing quantum-resistant key exchange algorithms, such as CRYSTALS-Kyber, provides secure key exchanges in encrypted communications, safeguarding data against the potential threats posed by quantum computing. For example, in wearable health monitoring devices, CRYSTALS-Kyber ensures that sensitive data, such as heart rate or glucose levels, is securely transmitted between devices and cloud services, protecting it from interception or tampering. By using quantum-resistant algorithms for key exchange, these devices

can maintain end-to-end security, even as quantum capabilities evolve, ensuring that personal health data remains protected and private in a connected ecosystem.

4.4 Standardization

PQC standardization is a critical step in preparing digital infrastructure for the advancement of quantum computing. In 2016, NIST launched a formal initiative to identify and standardize cryptographic algorithms resistant to the anticipated capabilities of quantum computers. This effort focuses on developing quantum-resistant algorithms to either complement or replace current standards, ensuring secure and resilient communication systems for the future. The standardization process is crucial for maintaining global cybersecurity, promoting interoperability, and encouraging widespread trust and adoption of quantum-resistant technologies. For the next-generation CE, PQC standardizations generally involve a multi-faceted approach. Key elements including to select suitable PQC algorithms for CE devices, requiring rigorous evaluation and performance testing, implementing seamless transitions, ensuring regulatory and compliance alignment, and fostering industry education. By addressing these components, secure and practical cryptographic systems can be developed to effectively counter the challenges posed by quantum computing advancements. Table 5 provides an overview of PQC algorithms selected for standardization along with key features, challenges, and current status.

4.5 Edge computing integration with PQC in CE devices

The edge computing technology integration with PQC and their implementation of CE devices can play a vital role in the efficient adaptation worldwide. CE devices can benefit from the edge computing facility by offloading computationally intensive cryptographic tasks to edge computing servers offered by 5G and B5G networks. Similarly, cloud computing backend systems can also be migrated to PQC-ready frameworks, using PQC schemes for secure cloud-to-edge communications and future-proofing cloud-based authentication, storage, and key management for CE devices.

5 Performance benchmarks and security metrics

The adoption of PQC for next-generation CE requires rigorous evaluation against existing cryptosystems. While RSA and ECC have historically provided efficiency and acceptable security in constrained devices, their reliance on number-theoretic assumptions renders them vulnerable to Shor's algorithm on scalable quantum computers. In contrast, PQC algorithms, particularly lattice-based schemes such as Kyber and Dilithium, must be benchmarked in terms of computational efficiency, communication cost, memory/energy consumption, and security metrics to determine their feasibility in next-generation CE.

5.1 Computational efficiency and latency

Benchmarks on high-performance servers indicate that PQC introduces a latency overhead of less than 5% compared to conventional cryptosystems, making it feasible for large-scale cloud and edge integration [140]. On constrained consumer devices, PQC execution time can vary by up to 12 across different algorithms and security levels [140, 141]. Specifically, lattice-based schemes such as Kyber and Dilithium demonstrate

Table 5 Overview of PQC algorithms selected for standardization along with key features, challenges, and current status

Algorithm	Type	Key features	Challenges	NIST status
ML-KEM (CRYSTALS–Kyber)	KEM	Ring-LWE problem, efficient polynomial operations, compact key sizes, FFT	Side-channel attack susceptibility, complex implementation, cryptanalysis resistance	Finalist
ML-DSA (CRYSTALS–Dilithium)	Digital signature	Module-lattice problems, efficient and fast signing/verification	Large signature sizes, side-channel vulnerabilities, cryptanalytic advances	Finalist
FN-DSA (FALCON)	Digital signature	NTRU lattice problem, compact signature sizes, FFT	Side-channel attack resistance, complexity of implementation, cryptanalysis resilience	Finalist
SLH-DSA (SPHINCS+)	Digital signature	Hash-based, stateless, no number-theoretic assumptions	Large signatures and public keys, resource intensity, implementation complexity	Finalist
Classic McEliece	KEM	Code-based, uses Goppa codes, long-term reliability	Large public key sizes, computational complexity, adoption resistance	Alternate
BIKE	KEM	Syndrome decoding, QC-MDPC codes, robustness	Implementation efficiency, side-channel resistance, cryptanalysis threats, error correction	Alternate
HQC	Public key encryption	Quasi-cyclic codes, strong security features	Performance and side-channel resistance	Selected for Standardization (March 2025)

efficient key encapsulation and signing operations, outperforming RSA-3072 by up to 35% in practical IoT and CE contexts [141]. Hardware-software co-design on RISC-V SoCs further enhances PQC performance, with optimized Dilithium implementations achieving up to a 5 speed-up compared to software-only baselines [142].

5.2 Communication and bandwidth overhead

Traditional ECC- and RSA-based TLS protocols rely on certificate management, increasing handshake sizes. PQC integration initially increases handshake payloads by up to 7; however, optimization strategies, including certificate elimination, hybrid key exchanges, and compression, can reduce this overhead by 40–60% [140]. In practice, certificate-free PQC schemes can reduce communication overhead by nearly 50%, decreasing handshake message sizes from 2.1 KB (ECC-TLS) to approximately 1.0 KB [141]. Such reductions are particularly important for consumer electronics operating under bandwidth-constrained 5G/6G networks and IoT environments.

5.3 Hardware throughput, memory, and energy

At the hardware level, PQC demonstrates strong performance. An ASIC implementation of the Kyber key encapsulation mechanism (KEM) in TSMC 40 nm technology achieved 877k operations per second (OPS), with an area efficiency of 0.276 kOPS/kGE and a power efficiency of 2272 kOPS/W [143]. By comparison, RSA-3072 achieves fewer than 200 OPS in equivalent hardware, highlighting a significant throughput advantage.

From a memory and energy perspective, lattice-based PQC schemes require 20–30% less memory than RSA at comparable security levels and reduce energy consumption by

up to 18% relative to ECC-256 [140]. These improvements directly contribute to longer battery life in mobile and embedded consumer electronics.

5.4 Security metrics

Unlike RSA-3072 (112-bit classical security) and ECC-256 (128-bit classical security), PQC schemes such as Kyber and Dilithium offer NIST Level 5 quantum-resilient security, equivalent to AES-256 symmetric strength [144]. This ensures robust protection against both Shor's and Grover's quantum attacks, providing long-term confidentiality for consumer applications, including mobile payments, biometric authentication, and secure IoT systems.

The key findings are summarized in Table 6. Representative graphs (Fig. 10, 11, 12) further illustrate the results: Figure 10 presents computational efficiency and latency across platforms, Fig. 11 highlights communication overhead in TLS handshakes, and Fig. 12 compares hardware throughput between conventional and post-quantum schemes

6 Challenges and future directions

In this section, we examine key challenges and future directions in areas that require concentrated efforts to strengthen CE security and privacy in a post-quantum world. Figure 13 highlights key challenges, such as resource constraints and deployment costs, along with future research areas including lightweight PQC algorithms, inter-industry collaboration, and policy development.

6.1 Challenges

- *Resource Constraints:* Implementing the PQC protocols on the next-generation CE is challenging because they are slightly computationally intensive, and most CE devices are generally equipped with limited computational resources. To tackle this, optimizing hardware and software is crucial to balance security with performance while designing PQC algorithms. Hence, lightweight PQC solutions, i.e., efficient

Table 6 Performance benchmarks of post-quantum cryptography compared with conventional cryptosystems

Metric	Post-quantum cryptography	Conventional cryptography (RSA/ECC)
Latency (server operations)	Less than or equal to 5% overhead in high-performance servers [140]	Baseline
Computational overhead (consumer devices)	Up to 12× variation across PQC schemes depending on algorithm and security level [140, 141]	Stable but vulnerable to quantum attacks
Communication overhead (TLS handshake)	Seven times raw increase, optimized to 40–60% reduction using certificate-free and hybrid schemes [141]	Elliptic Curve TLS requires 2.1 KB [141]
Key generation and encapsulation cycles	Kyber/Saber: 100,000–150,000 cycles [140]	RSA-3072: more than 500,000 cycles
Hardware throughput	Kyber ASIC in TSMC 40 nm achieves 877,000 operations per second with 2272 kOPS/W power efficiency [143]	RSA-3072 achieves fewer than 200 operations per second
Digital signature (Dilithium on RISC-V)	Up to 5× faster using hardware-software co-design compared to software-only implementation [142]	RSA/ECC implementations rely mainly on software
Memory and energy usage	20–30% less memory and 18% lower energy consumption [140]	Higher memory and energy consumption
Security level	AES-256 equivalent, NIST Level 5 quantum-resilient security [144]	ECC-256 provides only 128-bit classical security

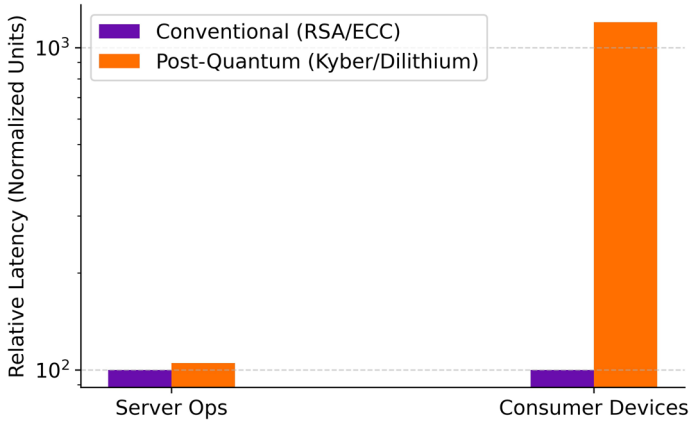

Fig. 10 Computational efficiency and latency

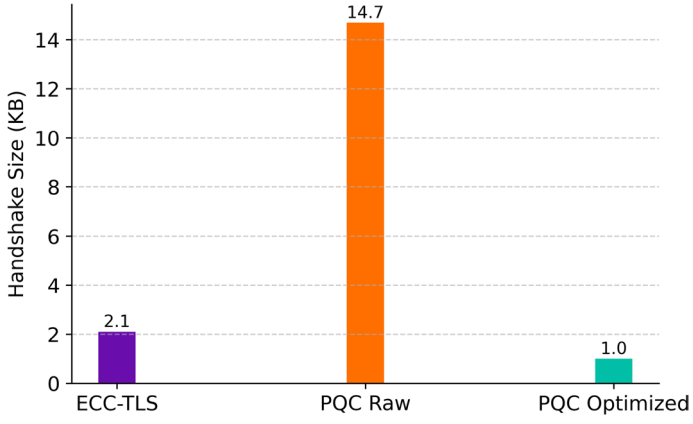

Fig. 11 Communication overhead in TLS handshake

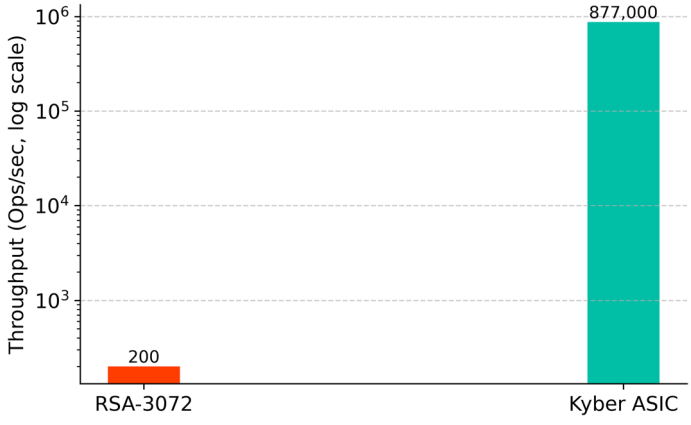

Fig. 12 Hardware throughput comparison

and secure, can play a significant role in this effort. Additionally, energy-efficient cryptographic accelerators can reduce power usage without compromising performance, making PQC practical for low-power devices like IoT gadgets and wearables.

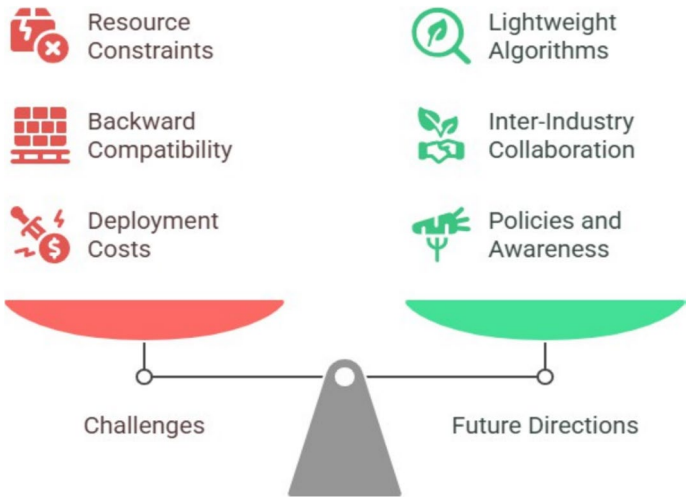

Fig. 13 Challenges and future research directions in PQC implementation for next-generation CE

- *Backward Compatibility:* Maintaining the interoperability of PQC schemes with existing classical cryptosystems is a critical challenge. To address this challenge, hybrid cryptosystems can provide an effective solution by combining classical and quantum-resistant algorithms. This ensures compatibility with current cryptographic infrastructures while paving the way for a gradual migration to PQC.
- *Deployment Costs:* The main challenge of deploying the PQC protocols on CE devices is due to the high costs of redesigning compatible devices and updating the overall infrastructure. However, this challenge can be addressed with the supportive and collaborative efforts of industry partnerships and public–private initiatives. Sharing resources and technical expertise will reduce deployment costs. Moreover, the support of governments and relevant international organizations can further ease the financial burden by providing subsidies or incentives to accelerate the adoption of quantum-safe solutions. Furthermore, implementing modular upgrade frameworks will allow for incremental deployment, enabling organizations to spread costs over time while progressively transitioning to PQC.

6.2 Future research directions

- *Lightweight Algorithms*: PQC is a newly emerging field that requires dedicated future research efforts to develop lightweight algorithms. Most existing PQC algorithms require high computational resources, which are not suitable for resource-constrained IoT and CE devices and sensors. Similarly, emerging technologies, such as neuromorphic computing and edge AI-based cryptographic accelerators, can be explored to improve the efficiency of existing PQC schemes and explore new ones. These innovations will make PQC protocols more practical and scalable for diverse CE applications.
- *Inter-Industry Collaboration*: Effective deployment of PQC protocols on CE devices requires close partnerships between semiconductor manufacturers, cryptography experts, the telecommunications sector, and CE companies. This collaboration will enable the co-design of hardware and software solutions optimized for seamless integration of PQC. Additionally, initiatives such as industry forums and consortia

can play a vital role in standardizing and accelerating PQC adoption across various CE segments, ensuring uniformity and broad industry alignment in the transition to quantum-resistant security for CE devices.

- *Policies and Awareness*: Policies at the government level need to be initiated to drive global collaboration, establish universally accepted PQC standards for CE devices, and address interoperability concerns. Public awareness campaigns about the importance of quantum-resistant security adaptation can foster consumer demand, further incentivizing manufacturers to adopt PQC standards.

7 Conclusions

Due to technological advancements, privacy and security challenges are more critical than ever in today's interconnected world. With the development of quantum computers, classical cryptographic methods that protect next-generation CE face significant vulnerabilities. One solution is to adopt PQC methods, which enhance security resilience against quantum attacks and are the most suitable methods for improving the security of next-generation CE devices. Given that prior literature has not specifically addressed the PQC solutions for CE. This article explored the importance, role, and benefits of PQC in securing next-generation CE. We analyzed the vulnerability of the classical cryptosystems in the context of quantum computers and discussed various PQC types. We discussed the implementation of PQC on next-generation CE. Finally, we explored the challenges and a few open research topics of PQC with the belief that these open research topics will help implement PQC solutions for next-generation CE.

Author contributions

Muhammad Asghar Khan contributed to the conceptualization, methodology, writing, reviewing, and editing. Fazal Noor supervised the research, and contributed to the revision of the manuscript. Justyna Zwiolek provided validation, resources, and funding acquisition, and was actively involved in writing, revising the manuscript, and addressing the reviewer's comments. Shumaila Javed contributed to writing, reviewing, editing, drafting the original manuscript and visualization. Each author has read and approved the final version of the manuscript.

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Declarations

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Consent for publication

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Competing interests

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References

1. Javeed D, Saeed MS, Ahmad I, Kumar P, Jolfaei A, Tahir M. An intelligent intrusion detection system for smart consumer electronics network. IEEE Trans Consum Electron. 2023;69(4):906–13.
2. Rani S, Gupta D, Garg S, Piran MJ, Hossain MS. Consumer electronic devices: evolution and edge security solutions. IEEE Consum Electron Mag. 2022;11(2):15–20.
3. Shruti, Rani S. Towards security of consumer electronic devices: an advanced attribute-based encryption approach for multimodal fog computing. IEEE Trans Consumer Electron. 2024, p. 1.
4. Sahu AK, Sahu M, Patro P, Sahu G, Nayak SR. Dual image-based reversible fragile watermarking scheme for tamper detection and localization. Pattern Anal Appl. 2022;26(2):571–90.

5. Patro P, Kumar K, Kumar GS, Sahu AK. Intelligent data classification using optimized fuzzy neural network and improved cuckoo search optimization. Iran J Fuzzy Syst. 2023;20(6):155–69.
6. Patro P, Kumar K, Suresh-Kumar G, Swain G. Similarity and wavelet transform based data partitioning and parameter learning for fuzzy neural network. J King Saud Univ Comput Inf Sci. 2022;34(6):3424–32.
7. Wang C, Li X, Xu H, Li Z, Wang J, Yang Z, et al. Towards practical quantum computers: transmon qubit with a lifetime approaching 0.5 milliseconds. NPJ Quantum Inf. 2022;8(1):3.
8. Yang Z, Zolanvari M, Jain R. A survey of important issues in quantum computing and communications. IEEE Commun Surv Tutor. 2023;25(2):1059–94.
9. Mulholland J, Mosca M, Braun J. The day the cryptography dies. IEEE Secur Priv. 2017;15(4):14–21.
10. Kaur J, Ramkumar K. The recent trends in cyber security: a review. J King Saud Univ Comput Inf Sci. 2022;34(8):5766–81.
11. Rivest RL, Shamir A, Adleman L. A method for obtaining digital signatures and public-key cryptosystems. Commun ACM. 1978;21(2):120–6.
12. Ramesh RK, Dodmane R, Shetty S, Aithal G, Sahu M, Sahu AK. A novel and secure fake-modulus based Rabin-cryptosystem. Cryptography. 2023;7(3):44.
13. Koblitz N. Elliptic curve cryptosystems. Math Comput. 1987;48(177):203–9.
14. Miller VS. Use of elliptic curves in cryptography. In: Conference on the theory and application of cryptographic techniques. Springer, 1985; p. 417–426.
15. Verma V. Lightweight mutual authentication protocol for iot devices using elliptical curves. Int J Electron Telecommun. 2024;70(4):785–90. https://doi.org/10.24425/ijet.2024.149609.
16. Diffie W, Hellman ME. New directions in cryptography. In: Democratizing cryptography: The Work of Whitfield Diffie and Martin Hellman, 2022, p. 365–390.
17. Fox D. Digital signature standard (dss). Datenschutz und Datensicherheit. 2001;25(6):186–92.
18. Shor PW. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. SIAM Rev. 1999;41(2):303–32.
19. Grover LK. A fast quantum mechanical algorithm for database search. In: Proceedings of the twenty-eighth annual ACM symposium on Theory of computing, 1996, p. 212–219.
20. Verma V. A study on quantum cryptography and its need. Stud Comput Intell. 2023;1085:407–35. https://doi.org/10.1007/978-981-19-9530-9_21.
21. Verma V, Malhotra K. A new secure quantum signature masked authentication scheme. Wirel Pers Commun. 2021;120:1659–74. https://doi.org/10.1007/s11277-021-08527-8.
22. Verma V. Unveiling the power of blockchain in pharmaceutical supply chains: strengthening security and improving drug traceability. El-Cezeri J Sci Eng. 2024;11(2):142–51.
23. Bernstein DJ. Introduction to post-quantum cryptography. In: Post-quantum cryptography. Springer, 2009; p. 1–14.
24. Bernstein DJ, Lange T. Post-quantum cryptography. Nature. 2017;549(7671):188–94.
25. Barreto PS, Piazza Biasi F, Dahab R, López-Hernández JC, de Morais EM, de Oliveira AD, Pereira GC, Ricardini JE. A panorama of post-quantum cryptography. In: Open problems in mathematics and computational science. 2014; p. 387–439.
26. Joseph D, Misoczki R, Manzano M, Tricot J, Pinuaga FD, Lacombe O, et al. Transitioning organizations to post-quantum cryptography. Nature. 2022;605(7909):237–43.
27. Khan MA, Khan MA. Securing IOT in the post-quantum era: implementation, challenges and future directions. IEEE Commun Standards Mag, 2025; p. 1
28. Yang Z, Alfauri H, Farkiani B, Jain R, D. Pietro R, Erbad A. A survey and comparison of post-quantum and quantum blockchains. IEEE Commun Surv Tutor. 2024;26(2):967–1002.
29. Or TRNGS, Adversaries AQ. Certifiable quantum dice. 2012.
30. Yang Z, Salman T, Jain R, Pietro RD. Decentralization using quantum blockchain: a theoretical analysis. IEEE Trans Quantum Eng. 2022;3:1–16.
31. Chowdhury K, Deb S, Roy KS, Podder D, Sahu AK. A systematic survey on TPE schemes for the cloud: classification, challenges, and future scopes. IEEE Access. 2024;12:160 669-160 689.
32. Wang X, Xu G, Yu Y. Lattice-based cryptography: a survey. Chin Ann Math Ser B. 2023;44(6):945–60.
33. Overbeck R, Sendrier N. Code-based cryptography. post-quantum cryptography. In: Bernstein DJ, Buchmann J, Dahmen E (eds). 2009.
34. Weger V, Gassner N, Rosenthal J. A survey on code-based cryptography, arXiv preprint arXiv:2201.07119, 2022.
35. Noel MD, Waziri VO, Abdulhamid SM, Ojeniyi JA. Review and analysis of classical algorithms and hash-based post-quantum algorithm. J Reliab Intell Environ. 2021; p. 1–18.
36. Ding J, Petzoldt A. Current state of multivariate cryptography. IEEE Secur Priv. 2017;15(4):28–36.
37. Peng C, Chen J, Zeadally S, He D. Isogeny-based cryptography: a promising post-quantum technique. IT Prof. 2019;21(6):27–32.
38. Kanwal S, Ali R. A cryptosystem with noncommutative platform groups. Neural Comput Appl. 2018;29:1273–8.
39. Sauerwein K, Butler JM, Reed C, Reczek KK. A NIST scientific foundation review. 2023.
40. Baldi M, Santini P, Cancellieri G. Post-quantum cryptography based on codes: state of the art and open challenges. In: 2017 AEIT international annual conference. IEEE, 2017; p. 1–6.
41. Mavroeidis V, Vishi K, Zych MD, Jøsang A. The impact of quantum computing on present cryptography, arXiv preprint arXiv:1804.00200, 2018.
42. Roy KS, Kalita HK. A survey on post-quantum cryptography for constrained devices. Int J Appl Eng Res. 2019;14(11):2608–15.
43. Clancy TC, McGwier RW, Chen L. Post-quantum cryptography and 5g security: tutorial. In: Proceedings of the 12th conference on security and privacy in wireless and mobile networks, 2019; p. 285–285.
44. Nejatollahi H, Dutt N, Ray S, Regazzoni F, Banerjee I, Cammarota R. Post-quantum lattice-based cryptography implementations: a survey. ACM Comput Surv. 2019;51(6):1–41.
45. Upadhyay L. Quantum cryptography: a survey. In: Innovations in bio-inspired computing and applications: proceedings of the 9th international conference on innovations in bio-inspired computing and applications (IBICA 2018) held in Kochi, India during December 17–19, 2018. Springer, 2019; pp. 20–35.

46. Fernandez-Carames TM, Fraga-Lamas P. Towards post-quantum blockchain: a review on blockchain cryptography resistant to quantum computing attacks. IEEE Access. 2020;8:21 091-21 116.
47. Kumar M, Pattnaik P. Post quantum cryptography(pqc)—an overview: (invited paper). In: 2020 IEEE high performance extreme computing conference (HPEC), 2020; p. 1–9.
48. Partala J. Post-quantum cryptography in 6g. In: 6G Mobile Wireless Networks, 2021; pp. 431–448.
49. Kan K, Une M et al. Recent trends on research and development of quantum computers and standardization of post-quantum cryptography. 2021.
50. Chamola V, Jolfaei A, Chanana V, Parashari P, Hassija V. Information security in the post quantum era for 5G and beyond networks: threats to existing cryptography, and post-quantum cryptography. Comput Commun. 2021;176:99–118.
51. Balamurugan C, Singh K, Ganesan G, Rajarajan M. Post-quantum and code-based cryptography—some prospective research directions. Cryptography. 2021;5(4):38.
52. Yalamuri G, Honnavalli P, Eswaran S. A review of the present cryptographic arsenal to deal with post-quantum threats. Proc Comput Sci. 2022;215:834–45.
53. Zeydan E, Turk Y, Aksoy B, Ozturk SB. Recent advances in post-quantum cryptography for networks: a survey. In: 2022 seventh international conference on mobile and secure services (MobiSecServ), 2022; p. 1–8.
54. Bavdekar R, Jayant Chopde E, Agrawal A, Bhatia A, Tiwari K. Post quantum cryptography: a review of techniques, challenges and standardizations. In: 2023 international conference on information networking (ICOIN), 2023; p. 146–151.
55. Sharma P, Gupta V, Sood SK. Post-quantum cryptography research landscape: a scientometric perspective. J Comput Inf Syst. 2023. https://doi.org/10.1080/08874417.2023.2260333.
56. Subramani S, Svn SK. Review of security methods based on classical cryptography and quantum cryptography. Cybernet Syst. 2023. https://doi.org/10.1080/01969722.2023.2166261.
57. Shaller A, Zamir L, Nojoumian M. Roadmap of post-quantum cryptography standardization: side-channel attacks and countermeasures. Inf Comput. 2023. https://doi.org/10.1016/j.ic.2023.105112.
58. Dam D-T, Tran T-H, Hoang V-P, Pham C-K, Hoang T-T. A survey of post-quantum cryptography: start of a new race. Cryptography. 2023;7(3):40.
59. Iqbal SS, Zafar A. A survey on post quantum cryptosystems: concept, attacks, and challenges in iot devices. In: 2023 10th international conference on computing for sustainable global development (INDIACom), 2023; p. 460–465.
60. Liu T, Ramachandran G, Jurdak R. Post-quantum cryptography for internet of things: a survey on performance and optimization, arXiv preprint arXiv:2401.17538, 2024.
61. Xie J, Zhao W, Lee H, Roy DB, Zhang X. Hardware circuits and systems design for post-quantum cryptography–a tutorial brief. IEEE Trans Circuits Syst II Express Briefs. 2024. https://doi.org/10.1109/TCSII.2024.3357836.
62. H. Gharavi, J. Granjal, and E. Monteiro, "Post-quantum blockchain security for the internet of things: Survey and research directions," *IEEE Communications Surveys Tutorials*, pp. 1–1, 2024.
63. Grote O, Ahrens A, Benavente-Peces C. A review of post-quantum cryptography and crypto-agility strategies. In: 2019 international interdisciplinary PhD workshop (IIPhDW), 2019; p. 115–120.
64. Khan MA, Javaid S, Mohsan SAH, Tanveer M, Ullah I. Future-proofing security for uavs with post-quantum cryptography: a review. IEEE Open J Commun Soc. 2024;5:6849–71.
65. Tranfield D, Denyer D, Smart P. Towards a methodology for developing evidence-informed management knowledge by means of systematic review. Br J Manage. 2003;14(3):207–22.
66. Khan AU, Javaid N, Khan MA, Ullah I. A blockchain scheme for authentication, data sharing and non-repudiation to secure internet of wireless sensor things. Cluster Comput. 2023;26(2):945–60. https://doi.org/10.1007/s10586-022-03722-z.
67. Khan S, Khan M, Khan MA, Khan MA, Wang L, Wu K. A blockchain-enabled ai-driven secure searchable encryption framework for medical IOT systems. IEEE J Biomed Health Inf. 2025; pp. 1–14
68. Yahaya AS, Javaid N, Ullah S, Khalid R, Javed MU, Khan RU, et al. A secure and efficient energy trading model using blockchain for a 5G-deployed smart community. Wirel Commun Mob Comput. 2022;2022:6953125. https://doi.org/10.1155/2022/6953125.
69. Nawaz SJ, Sharma SK, Patwary MN, Asaduzzaman M. Next-generation consumer electronics for 6G wireless era. IEEE Access. 2021;9:143 198-143 211.
70. Rupa C, Malleswari RP, Sultana SA, Abbas M, Sahu AK. Data privacy protection using lucas series based hybrid reversible watermarking approach. IEEE Access. 2024;12:134 578-134 593.
71. Podder D, Deb S, Banik D, et al. Robust medical and color image cryptosystem using array index and chaotic s-box. Cluster Comput. 2024;27:4321–46. https://doi.org/10.1007/s10586-024-04584-3.
72. Sahu AK, Sahu M, Patro P, Sahu G, Nayak SR. Dual image-based reversible fragile watermarking scheme for tamper detection and localization. Pattern Anal Appl. 2023;26:571–90. https://doi.org/10.1007/s10044-022-01104-0.
73. Nawaz SJ, Sharma SK, Wyne S, Patwary MN, Asaduzzaman M. Quantum machine learning for 6G communication networks: state-of-the-art and vision for the future. IEEE Access. 2019;7:46 317-46 350.
74. Deutsch D. Quantum theory, the church–turing principle and the universal quantum computer. Proc R Soc Lond Math Phys Sci. 1985;100(1818):97–117.
75. Clarke J, Wilhelm FK. Superconducting quantum bits. Nature. 2008;453(7198):1031–42.
76. Marella ST, Parisa HSK. Introduction to quantum computing. Quantum Computing and Communications, 2020.
77. Riel H. Quantum computing technology. In: 2021 IEEE International Electron Devices Meeting (IEDM). IEEE, 2021; p. 1–3.
78. Mo¨ller M, Vuik C. On the impact of quantum computing technology on future developments in high-performance scientific computing. Ethics Inf Technol. 2017;19:253–69.
79. Kitaev AY, Shen A, Vyalyi MN. Classical and quantum computation. New York: American Mathematical Society; 2002.
80. Nielsen MA, Chuang IL. Quantum computation and quantum information. Phys Today. 2001;54(2):60.
81. Aaronson S. The limits of quantum. Sci Am. 2008;298(3):62–9.
82. Seo WY. Comparing RSA ECC and post quantum cryptography. J Math Anal Appl. 2018;10:19–33.
83. Gajbhiye S, Karmakar S, Sharma M, Sharma S. Paradigm shift from classical cryptography to quantum cryptography. In: 2017 international conference on intelligent sustainable systems (ICISS). IEEE, 2017; pp. 548–555.
84. Ahilan A, Jeyam A. Breaking barriers in conventional cryptography by integrating with quantum key distribution. Wirel Pers Commun. 2023;129(1):549–67.

85. Shiu H-J, Yang C-T, Tsai Y-R, Lin W-C, Lai C-M. Maintaining secure level on symmetric encryption under quantum attack. Appl Sci. 2023;13(11):6734.
86. Shrivastava P, Soni KK, Rasool A. Evolution of quantum computing based on grover's search algorithm. In: 2019 10th international conference on computing, communication and networking technologies (ICCCNT). IEEE, 2019; pp. 1–6.
87. Sahu M, Padhy N, Gantayat SS, Sahu AK. Shadow image based reversible data hiding using addition and subtraction logic on the lsb planes. Sens Imaging. 2021;22(1):7.
88. [Online]. Available: https://www.accenture.com/ar-es/services/accenture-labs/cybersecurity
89. Cavaliere F, Mattsson J, Smeets B. The security implications of quantum cryptography and quantum computing. Netw Secur. 2020;2020(9):9–15.
90. Chuang C, Hsu I, Lee T. A generalized distributed RSA key generation. Cryptology ePrint Archive, 2024.
91. Rotem L, Segev G. Generically speeding-up repeated squaring is equivalent to factoring: Sharp thresholds for all generic-ring delay functions. In: Annual international cryptology conference. Springer, 2020; p. 481–509.
92. Benger N. Cryptographic pairings: efficiency and DLP security, Ph.D. dissertation, Dublin City University, 2010.
93. Ullah I, Amin NU, Almogren A, Khan MA, Uddin MI, Hua Q. A lightweight and secured certificate-based proxy signcryption (cb-ps) scheme for e-prescription systems. IEEE Access. 2020;8:199 197-199 212.
94. Suo J, Wang L, Yang S, Zheng W, Zhang J. Quantum algorithms for typical hard problems: a perspective of cryptanalysis. Quantum Inf Process. 2020;19:1–26.
95. Dyakonov M. When will useful quantum computers be constructed? Not in the foreseeable future, this physicist argues. Here's why: the case against: quantum computing. IEEE Spectr. 2019;56(3):24–9.
96. Zhou B-M, Yuan Z. Breaking symmetric cryptosystems using the offline distributed grover-meets-simon algorithm. Quantum Inf Process. 2023;22(9):333.
97. Langenberg BW. Quantum circuits for symmetric cryptanalysis, Ph.D. dissertation, Florida Atlantic University, 2018.
98. Jordan SP, Liu Y-K. Quantum cryptanalysis: Shor, Grover, and beyond. IEEE Secur Priv. 2018;16(5):14–21.
99. Isa H, Bahari I, Sufian H, Z'aba MR, Aes: current security and efficiency analysis of its alternatives. In: 2011 7th international conference on information assurance and security (IAS). IEEE. 2011; pp. 267–274.
100. Alhag NMM, Mohamed YA. An enhancement of data encryption standards algorithm (des). In: 2018 international conference on computer, control, electrical, and electronics engineering (ICCCEEE), 2018; p. 1–6.
101. Bernstein DJ, The salsa20 family of stream ciphers. In: New stream cipher designs: the eSTREAM finalists. Springer, 2008; p. 84–97.
102. McGrew DA, Viega J. The security and performance of the galois/counter mode (gcm) of operation. In: International conference on cryptology in India. Springer, 2004; p. 343–355.
103. Bernstein DJ. The poly1305-aes message-authentication code. In: International workshop on fast software encryption. Springer, 2005; p. 32–49.
104. Gilbert H, Handschuh H. Security analysis of sha-256 and sisters. In: International workshop on selected areas in cryptography. Springer, 2003; p. 175–193.
105. Dworkin MJ. Sha-3 standard: permutation-based hash and extendable-output functions, 2015.
106. Imam R, Areeb QM, Alturki A, Anwer F. Systematic and critical review of RSA based public key cryptographic schemes: past and present status. IEEE Access. 2021;9:155 949-155 976.
107. Kivinen T, Kojo M. More modular exponential (modp) Diffie–Hellman groups for internet key exchange (ike). Tech Rep. 2003.
108. Goldwasser S, Micali S, Rivest RL. A digital signature scheme secure against adaptive chosen-message attacks. SIAM J Comput. 1988;17(2):281–308.
109. Cohen H, Frey G, Avanzi R, Doche C, Lange T, Nguyen K, et al. Handbook of elliptic and hyperelliptic curve cryptography. Boca Raton: CRC Press; 2005.
110. Johnson D, Menezes A, Vanstone S. The elliptic curve digital signature algorithm (ecdsa). Int J Inf Secur. 2001;1:36–63.
111. N. Farrugia, D. Bonanno, N. Frendo, and A. Xuereb, "Pqc and qkd are both required to enable a quantum-safe future," in *Toward a Quantum-Safe Communication Infrastructure*. IOS Press, 2024, pp. 24–36.
112. Cherbal S, Zier A, Hebal S, Louail L, Annane B. Security in internet of things: a review on approaches based on blockchain, machine learning, cryptography, and quantum computing. J Supercomput. 2024;80(3):3738–816.
113. Alnahawi N, Wiesmaier A, Grasmeyer T, Geißler J, Zeier A, Bauspieß A, Heinemann A. On the state of post-quantum cryptography migration. 2021.
114. Khan MA, Kumar N, Alsamhi SH, Barb G, Zywiołek J, Ullah I, et al. Security and privacy issues and solutions for UAVs in B5G networks: a review. IEEE Trans Netw Serv Manag. 2025;22(1):892–912.
115. Joshi A, Bairwa AK, Pljonkin AP, Garg P, Agrawal K. From pre-quantum to post-quantum rsa. In: Proceedings of the 6th international conference on networking, intelligent systems & security. 2023; p. 1–8.
116. Overbeck R, Sendrier N. Code-based cryptography. In: Post-quantum cryptography. Springer, 2009; p. 95–145.
117. Buchmann J, Dahmen E, Szydlo M. Hash-based digital signature schemes. In: Post-quantum cryptography. Springer, 2009; p. 35–93.
118. Wolf C. Multivariate quadratic polynomials in public key cryptography. Cryptology ePrint Archive. 2005.
119. Costello C, Longa P, Naehrig M. Efficient algorithms for supersingular isogeny Diffie–Hellman. In: Advances in cryptology—CRYPTO 2016: 36th annual international cryptology conference, Santa Barbara, CA, USA, August 14–18, 2016, Proceedings, Part I 36. Springer, 2016; p. 572–601.
120. Myasnikov AG, Shpilrain V, Ushakov A. Non-commutative cryptography and complexity of group-theoretic problems. New York: American Mathematical Society; 2011.
121. Micciancio D, Regev O. Lattice-based cryptography. In: Post-quantum cryptography. Springer, 2009, p. 147–191.
122. Ding J, Lindner R. Identifying ideal lattices. Cryptology ePrint Archive. 2007.
123. Chi DP, Choi JW, San Kim J, Kim T. Lattice based cryptography for beginners. Cryptology ePrint Archive, 2015.
124. Nguyen PQ, Stern J. Lattice reduction in cryptology: an update. In: International algorithmic number theory symposium. Springer, 2000; p. 85–112.
125. Li Y, Ng KS, Purcell M. A tutorial introduction to lattice-based cryptography and homomorphic encryption, arXiv preprint arXiv:2208.08125, 2022.
126. Yu Y. Preface to special topic on lattice-based cryptography. Natl Sci Rev. 2021;8(9):154.

127. Bandara H, Herath Y, Weerasundara T, Alawatugoda J. On advances of lattice-based cryptographic schemes and their implementations. Cryptography. 2022;6(4):56.
128. Mariano A, Laarhoven T, Correia F, Rodrigues M, Falcao G. A practical view of the state-of-the-art of lattice-based cryptanalysis. IEEE Access. 2017;5:24 184-24 202.
129. Brijwani GN, Ajmire PE, Thawani PV. Future of quantum computing in cyber security. In: Handbook of research on quantum computing for smart environments. IGI Global, 2023; p. 267–298.
130. Sendrier N. Code-based cryptography: state of the art and perspectives. IEEE Secur Priv. 2017;15(4):44–50.
131. Butin D. Hash-based signatures: state of play. IEEE Secur Priv. 2017;15(4):37–43.
132. Gheorghiu V, Mosca M. Benchmarking the quantum cryptanalysis of symmetric, public-key and hash-based cryptographic schemes, arXiv preprint arXiv:1902.02332, 2019.
133. Menezes AJ, Van Oorschot PC, Vanstone SA. Handbook of applied cryptography. Boca Raton: CRC Press; 2018.
134. Merkle RC. A certified digital signature. In: Conference on the theory and application of cryptology. Springer, 1989; p. 218–238.
135. Merkle RC. One way hash functions and des. In: Conference on the theory and application of cryptology. Springer, 1989; p. 428–446.
136. Damga˚rd IB, A design principle for hash functions. In: Conference on the theory and application of cryptology. Springer, 1989, p. 416–427.
137. Merkle RC. A digital signature based on a conventional encryption function. In: Conference on the theory and application of cryptographic techniques. Springer, 1987; p. 369–378.
138. Greconici DO, Kannwischer MJ, Sprenkels A. Compact dilithium implementations on cortex-m3 and cortex-m4. IACR transactions on cryptographic hardware and embedded systems, 2021; p. 1–24.
139. Amazon Web Services. How to tune tls for hybrid post-quantum cryptography with kyber, https://aws.amazon.com/blogs/security/how-to-tune-tls-for-hybrid-post-quantum-cryptography-with-kyber/, 2023. Accessed: 2025-06-19.
140. Abbasi M, Cardoso F, Váz P, Silva J, Martins P. A practical performance benchmark of post-quantum cryptography across heterogeneous computing environments. Cryptography. 2025;9(2):32.
141. Commey D, Appiah B, Klogo GS, Bagyl-Bac W, Gadze JD. Performance analysis and deployment considerations of post-quantum cryptography for consumer electronics, arXiv preprint arXiv:2505.02239, May 2025, available at: https://arxiv.org/abs/2505.02239.
142. Wang T, Zhang C, Zhang X, Gu D, Cao P. Optimized hardware–software co-design for kyber and dilithium on risc-v soc fpga. IACR Trans Cryptogr Hardw Embed Syst. 2024;2024(3):99–135.
143. Wang T, Zhang C, Zhang X, Gu D, Cao P. High-throughput ASIC implementation of kyber post-quantum key encapsulation mechanism. Electronics. 2024;14(15):2969.
144. NIST PQC Standardization Project. Post-quantum cryptography security levels, https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization, 2024.

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